

Tangent and Cotangent Bundles

Special Mathematics Lectures - Differential Geometry

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1 Tangent bundles

Consider a n -dimensional smooth manifold M . This manifold is equipped with a maximal smooth atlas $\mathcal{U} = (U_\alpha, \varphi_\alpha)_\alpha$. For each point $p \in M$, there is a tangent space $T_p(M)$, which contains all tangent vectors $X_p : C^\infty(p) \rightarrow \mathbb{R}$. Given a local coordinate system (U, φ) such that $p \in U$, every tangent vector X_p can be expressed as

$$X_p = \sum_{i=1}^n \alpha^i E_{i,p} = \sum_{i=1}^n \alpha^i (\varphi^{-1})_* \left(\frac{\partial}{\partial x^i} \Big|_{\varphi(p)} \right),$$

for some coefficients $\alpha^1, \dots, \alpha^n \in \mathbb{R}$. Furthermore, we have

$$\begin{aligned} \varphi_*(X_p) &= \varphi_* \left[\sum_{i=1}^n \alpha^i (\varphi^{-1})_* \left(\frac{\partial}{\partial x^i} \Big|_{\varphi(p)} \right) \right] = \sum_{i=1}^n \alpha^i [\varphi_* \circ (\varphi^{-1})_*] \left(\frac{\partial}{\partial x^i} \Big|_{\varphi(p)} \right) \\ &= \sum_{i=1}^n \alpha^i (\varphi \circ \varphi^{-1})_* \left(\frac{\partial}{\partial x^i} \Big|_{\varphi(p)} \right) \\ &= \sum_{i=1}^n \alpha^i \frac{\partial}{\partial x^i} \Big|_{\varphi(p)} \end{aligned}$$

for any tangent vector $X_p \in T_p(M)$, which shows that $\varphi_*(X_p) \in T_{\varphi(p)}(\mathbb{R}^n)$. Since there is an one-to-one correspondence between elements of $T_{\varphi(p)}(\mathbb{R}^n)$ and elements of $\mathbb{R}^n$, we can identify $\varphi_*(X_p)$ with $(\alpha^1, \dots, \alpha^n) \in \mathbb{R}^n$. Or more properly, by using the map $c : T_{\varphi(p)}(\mathbb{R}^n) \rightarrow \mathbb{R}^n$ given by

$$c \left(\sum_{i=1}^n \alpha^i \frac{\partial}{\partial x^i} \Big|_{\varphi(p)} \right) = (\alpha^1, \dots, \alpha^n),$$

we can have $c \circ \varphi_*(X_p) = (\alpha^1, \dots, \alpha^n)$. Now, the tangent bundle $T(M)$ of the manifold M is defined by

$$T(M) := \bigsqcup_{p \in M} T_p(M).$$

We want to construct a differential structure on this object. Before doing that, we want to introduce a natural projection $\pi : T(M) \rightarrow M$ defined by $\pi(X_p) = p$ for any element $X_p \in T(M)$. Also, for any point $p \in M$, the tangent space $T_p(M)$ is the preimage of $\{p\}$ under the map π , i.e. $\pi^{-1}(\{p\}) = T_p(M)$, which is also called the *fiber* at p . Consider a local coordinate system (U, φ) in the maximal smooth atlas $\mathcal{U}$ on M . Then, we define the subset $\tilde{U} = \pi^{-1}(U)$ of $T(M)$ and a map $\tilde{\varphi} : \tilde{U} \rightarrow \mathbb{R}^{2n}$, which is given by

$$\tilde{\varphi}(X) = (\varphi \circ \pi(X), c \circ \varphi_*(X)) = (x^1, \dots, x^n, \alpha^1, \dots, \alpha^n),$$

for any tangent vector $X \in \tilde{U}$ with its projection $\pi(X) = p \in U$. This map is bijective and its inverse $\tilde{\varphi}^{-1} : \tilde{\varphi}(\tilde{U}) \rightarrow \tilde{U}$ is given by

$$\tilde{\varphi}^{-1}(x^1, \dots, x^n, \alpha^1, \dots, \alpha^n) = \sum_{i=1}^n \alpha^i E_{i, \varphi^{-1}(x^1, \dots, x^n)} = \sum_{i=1}^n \alpha^i (\varphi^{-1})_* \left(\frac{\partial}{\partial x^i} \Big|_{(x^1, \dots, x^n)} \right),$$

for any $(x^1, \dots, x^n, \alpha^1, \dots, \alpha^n) \in \tilde{\varphi}(\tilde{U}) \subset \mathbb{R}^{2n}$. The following lemma shows that all of the pairs $(\tilde{U}, \tilde{\varphi})$ are C^∞ compatible.

Lemma 1. *If the pair $(\tilde{U}_\alpha, \tilde{\varphi}_\alpha)$ and $(\tilde{U}_\beta, \tilde{\varphi}_\beta)$ corresponds to the two coordinate systems $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) , then the two maps*

$$\tilde{\varphi}_\alpha \circ \tilde{\varphi}_\beta^{-1} : \tilde{\varphi}_\beta(\tilde{U}_\alpha \cap \tilde{U}_\beta) \rightarrow \tilde{\varphi}_\alpha(\tilde{U}_\alpha \cap \tilde{U}_\beta),$$

and

$$\tilde{\varphi}_\beta \circ \tilde{\varphi}_\alpha^{-1} : \tilde{\varphi}_\alpha(\tilde{U}_\alpha \cap \tilde{U}_\beta) \rightarrow \tilde{\varphi}_\beta(\tilde{U}_\alpha \cap \tilde{U}_\beta)$$

are smooth maps.

Proof. First, consider any point $p \in U_\alpha \cap U_\beta$ and any smooth function $f \in C^\infty(p)$. Then, we have for $1 \leq i \leq n$,

$$\begin{aligned} E_{i,p}^\beta(f) &= (\varphi_\beta^{-1})_* \left(\frac{\partial}{\partial x^i} \Big|_{\varphi_\beta(p)} \right) (f) = \frac{\partial}{\partial x^i} (f \circ \varphi_\beta^{-1}) \Big|_{\varphi_\beta(p)} \\ &= \frac{\partial}{\partial x^i} (f \circ \varphi_\alpha^{-1} \circ \varphi_\alpha \circ \varphi_\beta^{-1}) \Big|_{\varphi_\beta(p)} \\ &= \sum_{j=1}^n \frac{\partial}{\partial x^i} (\varphi_\alpha \circ \varphi_\beta^{-1})^j \Big|_{\varphi_\beta(p)} \frac{\partial}{\partial x^j} (f \circ \varphi_\alpha^{-1}) \Big|_{\varphi_\alpha(p)} \\ &= \sum_{j=1}^n \frac{\partial}{\partial x^i} (\varphi_\alpha \circ \varphi_\beta^{-1})^j \Big|_{\varphi_\beta(p)} E_{j,p}^\alpha(f), \end{aligned}$$

where

$$E_{j,p}^\alpha = (\varphi_\alpha^{-1})_* \left(\frac{\partial}{\partial x^j} \Big|_{\varphi_\alpha(p)} \right).$$

Hence,

$$E_{i,p}^\beta = \sum_{j=1}^n \frac{\partial}{\partial x^i} (\varphi_\alpha \circ \varphi_\beta^{-1})^j \Big|_{\varphi_\beta(p)} E_{j,p}^\alpha.$$

Thus, for any $(\beta^1, \dots, \beta^n) \in \mathbb{R}^n$, we have

$$\begin{aligned} \tilde{\varphi}_\alpha \circ \tilde{\varphi}_\beta^{-1} (\varphi_\beta(p), \beta^1, \dots, \beta^n) &= \tilde{\varphi}_\alpha \left(\sum_{i=1}^n \beta^i E_{i,p}^\beta \right) = \tilde{\varphi}_\alpha \left(\sum_{j=1}^n \alpha^j E_{j,p}^\alpha \right) \\ &= (\varphi_\alpha \circ \varphi_\beta^{-1} (\varphi_\beta(p)), \alpha^1, \dots, \alpha^n), \end{aligned}$$

where

$$\alpha^j := \sum_{i=1}^n \beta^i \frac{\partial}{\partial x^i} (\varphi_\alpha \circ \varphi_\beta^{-1})^j \Big|_{\varphi_\beta(p)}.$$

Therefore, since $\varphi_\alpha \circ \varphi_\beta^{-1}$ and all of its derivatives are smooth maps, the map $\tilde{\varphi}_\alpha \circ \tilde{\varphi}_\beta^{-1}$ is also a smooth map. A similar conclusion can be also applied to $\tilde{\varphi}_\beta \circ \tilde{\varphi}_\alpha^{-1}$ by using the fact that $\varphi_\beta \circ \varphi_\alpha^{-1}$ is also smooth. $\square$

Using all pairs $(\tilde{U}, \tilde{\varphi})$, which correspond to all coordinate systems (U, φ) in the maximal smooth atlas $\mathcal{U}$ on M , and the usual topology of $\mathbb{R}^{2n}$, we construct a basis $\mathcal{B}$ of a topology on $T(M)$. We define $\tilde{\mathcal{B}}$ as a collection of all sets $A \subset T(M)$ such that there exists a local coordinate system (U, φ) , so that $A \subset \tilde{U}$ and the set $\tilde{\varphi}(A)$ is an open subset of $\mathbb{R}^{2n}$. With the maximal smooth atlas $\mathcal{U} = (U_\alpha, \varphi_\alpha)_\alpha$, this collection can be expressed as

$$\tilde{\mathcal{B}} = \bigcup_{\alpha} \left\{ A \in \mathcal{P}(T(M)) \mid A \subset \tilde{U}_{\alpha}, \text{ and } \tilde{\varphi}_{\alpha}(A) \text{ is open in } \mathbb{R}^{2n} \right\}.$$

The following lemma shows that $\tilde{\mathcal{B}}$ satisfies the conditions (Proposition A3) to be a basis for some topology on $T(M)$.

Lemma 2. *The collection $\tilde{\mathcal{B}}$ of subsets of $T(M)$ satisfies*

- i) *The tangent bundle $T(M)$ is the union of all sets $A \in \tilde{\mathcal{B}}$.*
- ii) *For any two sets A and B in $\tilde{\mathcal{B}}$, their intersection $A \cap B$ also belongs to $\tilde{\mathcal{B}}$.*

Proof. i) Let $\mathcal{U} = (U_{\alpha}, \varphi_{\alpha})$ be the maximal smooth atlas for M . For any α , the set

$$\begin{aligned} \tilde{\varphi}_{\alpha}(\tilde{U}_{\alpha}) &= \tilde{\varphi}_{\alpha}(\pi^{-1}(U_{\alpha})) = \tilde{\varphi}_{\alpha} \left(\bigsqcup_{p \in U_{\alpha}} T_p(M) \right) = \bigcup_{p \in U_{\alpha}} \tilde{\varphi}_{\alpha}(T_p(M)) = \bigcup_{p \in U_{\alpha}} \{\varphi_{\alpha}(p)\} \times \mathbb{R}^n \\ &= \varphi_{\alpha}(U_{\alpha}) \times \mathbb{R}^n \subset \mathbb{R}^{2n} \end{aligned}$$

is open in $\mathbb{R}^{2n}$ because φ_{α} is bi-continuous and hence, $\varphi_{\alpha}(U_{\alpha})$ is open in $\mathbb{R}^n$. Therefore, we have $\tilde{U}_{\alpha} \in \tilde{\mathcal{B}}$ for any α . It follows that

$$T(M) = \pi^{-1}(M) = \pi^{-1} \left(\bigcup_{\alpha} U_{\alpha} \right) = \bigcup_{\alpha} \pi^{-1}(U_{\alpha}) = \bigcup_{\alpha} \tilde{U}_{\alpha} \subset \bigcup_{A \in \tilde{\mathcal{B}}} A.$$

Since $A \subset T(M)$ for any $A \in \tilde{\mathcal{B}}$, we have

$$\left(\bigcup_{A \in \tilde{\mathcal{B}}} A \right) \subset T(M).$$

Therefore, $T(M)$ is the union of all sets in $\tilde{\mathcal{B}}$, that is $T(M) = \bigcup_{A \in \tilde{\mathcal{B}}} A$.

- ii) For any two sets A and B in $\tilde{\mathcal{B}}$, there exist two coordinate systems $(U_{\alpha}, \varphi_{\alpha})$ and $(U_{\beta}, \varphi_{\beta})$ such that $A \subset \tilde{U}_{\alpha}$, $B \subset \tilde{U}_{\beta}$, and $\tilde{\varphi}_{\alpha}(A)$, $\tilde{\varphi}_{\beta}(B)$ are open sets in $\mathbb{R}^{2n}$. First, we have

$$\tilde{\varphi}_{\alpha}(\tilde{U}_{\alpha} \cap \tilde{U}_{\beta}) = \tilde{\varphi}_{\alpha}(\pi^{-1}(U_{\alpha}) \cap \pi^{-1}(U_{\beta})) = \tilde{\varphi}_{\alpha}(\pi^{-1}(U_{\alpha} \cap U_{\beta})) = \varphi_{\alpha}(U_{\alpha} \cap U_{\beta}) \times \mathbb{R}^n,$$

where the last equality is due to the similarity with $\tilde{\varphi}_{\alpha}(\tilde{U}_{\alpha})$ above. Then, $\tilde{\varphi}_{\alpha}(\tilde{U}_{\alpha} \cap \tilde{U}_{\beta})$ is an open set of $\mathbb{R}^{2n}$ because $\varphi_{\alpha}(U_{\alpha} \cap U_{\beta})$ is open in $\mathbb{R}^n$. Similarly, $\tilde{\varphi}_{\beta}(\tilde{U}_{\alpha} \cap \tilde{U}_{\beta})$ is also open in $\mathbb{R}^{2n}$. Hence, the sets

$$\tilde{\varphi}_{\alpha}(A \cap (\tilde{U}_{\alpha} \cap \tilde{U}_{\beta})) = \tilde{\varphi}_{\alpha}(A) \cap \tilde{\varphi}_{\alpha}(\tilde{U}_{\alpha} \cap \tilde{U}_{\beta}),$$

and

$$\tilde{\varphi}_{\beta}(B \cap (\tilde{U}_{\alpha} \cap \tilde{U}_{\beta})) = \tilde{\varphi}_{\beta}(B) \cap \tilde{\varphi}_{\beta}(\tilde{U}_{\alpha} \cap \tilde{U}_{\beta})$$

are open sets in $\mathbb{R}^{2n}$ because they are intersections of open sets. Note that the image of intersection is the intersection of images because $\tilde{\varphi}_{\alpha}$ and $\tilde{\varphi}_{\beta}$ are bijective. In addition, since the transition map $\tilde{\varphi}_{\alpha} \circ \tilde{\varphi}_{\beta}^{-1}$ is bi-continuous (Lemma 1), the set

$$\tilde{\varphi}_{\alpha}(B \cap (\tilde{U}_{\alpha} \cap \tilde{U}_{\beta})) = \tilde{\varphi}_{\alpha} \circ \tilde{\varphi}_{\beta}^{-1} \left(\tilde{\varphi}_{\beta}(B \cap (\tilde{U}_{\alpha} \cap \tilde{U}_{\beta})) \right)$$

is also open in $\mathbb{R}^{2n}$. Therefore, since $A \cap B \subset \tilde{U}_{\alpha} \cap \tilde{U}_{\beta}$, we have

$$A \cap B = A \cap B \cap (\tilde{U}_{\alpha} \cap \tilde{U}_{\beta}) = (A \cap (\tilde{U}_{\alpha} \cap \tilde{U}_{\beta})) \cap (B \cap (\tilde{U}_{\alpha} \cap \tilde{U}_{\beta}))$$

$$\Rightarrow \quad \tilde{\varphi}_\alpha(A \cap B) = \tilde{\varphi}_\alpha\left(A \cap (\tilde{U}_\alpha \cap \tilde{U}_\beta)\right) \cap \tilde{\varphi}_\alpha\left(B \cap (\tilde{U}_\alpha \cap \tilde{U}_\beta)\right),$$

which implies that $\tilde{\varphi}_\alpha(A \cap B)$ is an open set of $\mathbb{R}^{2n}$. Now, we define a new coordinate system $(U_\gamma, \varphi_\gamma)$, where $U_\gamma = U_\alpha \cap U_\beta$ and $\varphi_\gamma = \varphi_\alpha|_{U_\gamma}$. The smooth atlas is maximal, so it also contains this coordinate system. This coordinate system generates a new pair $(\tilde{U}_\gamma, \tilde{\varphi}_\gamma)$. Also, $A \cap B \subset U_\gamma$, and $\tilde{\varphi}_\gamma(A \cap B) = \tilde{\varphi}_\alpha(A \cap B)$ is open in $\mathbb{R}^{2n}$. As a result, $A \cap B \in \tilde{\mathcal{B}}$. $\square$

Remark. We have demonstrated that the collection of open sets $\{\tilde{U}_\alpha\}_\alpha$ covers the tangent bundle $T(M)$. Consequently, the pair $(\tilde{U}_\alpha, \tilde{\varphi}_\alpha)$ meets the two main conditions (covering and C^∞ -compatibility) necessary to constitute a smooth atlas on $T(M)$ after constructing a topology on $T(M)$ and verifying that it is indeed a topological manifold.

With the basis $\tilde{\mathcal{B}}$, a topology $\tilde{\mathcal{T}}$ of $T(M)$ is defined as the collection of all unions of subsets in $\tilde{\mathcal{B}}$ (Proposition A3). In other words, any open set A of $T(M)$ can be expressed as $A = \bigcup_\beta \tilde{B}_\beta$ for some subfamily $\{\tilde{B}_\beta\} \subset \tilde{\mathcal{B}}$. Another way to determine if a subset of $T(M)$ is open is shown in the following lemma.

Lemma 3. *A subset A of $T(M)$ is an open subset if and only if $\tilde{\varphi}_\alpha(A \cap \tilde{U}_\alpha)$ is open in $\mathbb{R}^{2n}$ for any α .*

Proof. Suppose that A is an open subset of $T(M)$. Then, there exists a subfamily $\{\tilde{B}_\alpha\} \subset \tilde{\mathcal{B}}$ such that $\tilde{B}_\alpha \subset \tilde{U}_\alpha$ and $A = \bigcup_\alpha \tilde{B}_\alpha$. Note that $\tilde{B}_\alpha$ can be empty for some α . Then, we have

$$A \cap \tilde{U}_\beta = \left(\bigcup_\alpha \tilde{B}_\alpha \right) \cap \tilde{U}_\beta = \bigcup_\alpha \tilde{B}_\alpha \cap \tilde{U}_\beta = \tilde{B}_\beta$$

because $\tilde{B}_\alpha \cap \tilde{U}_\beta = \emptyset$ if $\alpha \neq \beta$ and $\tilde{B}_\alpha \cap \tilde{U}_\beta = \tilde{B}_\beta$ if $\alpha = \beta$. In addition, since $\tilde{B}_\alpha \in \tilde{\mathcal{B}}$, the set $\tilde{\varphi}_\alpha(\tilde{B}_\alpha)$ is an open subset of $\mathbb{R}^{2n}$. Consequently, the set $\tilde{\varphi}_\alpha(A \cap \tilde{U}_\alpha)$ is an open subset of $\mathbb{R}^{2n}$.

Conversely, suppose that $\tilde{\varphi}_\alpha(A \cap \tilde{U}_\alpha)$ is an open subset of $\mathbb{R}^{2n}$ for every α . Therefore, the subset $\tilde{B}_\alpha := A \cap \tilde{U}_\alpha \subset \tilde{U}_\alpha$ belongs to $\tilde{\mathcal{B}}$ for every α . As a result,

$$\bigcup_\alpha \tilde{B}_\alpha = \bigcup_\alpha A \cap \tilde{U}_\alpha = A \cap \left(\bigcup_\alpha \tilde{U}_\alpha \right) = A \cap T(M) = A,$$

which implies that A is an open subset of $T(M)$. $\square$

Having constructed a topology on $T(M)$, the next task is to prove that $T(M)$ is a smooth manifold, which is done in the following statements.

Proposition 4. *The tangent bundle $T(M)$ with the topology $\tilde{\mathcal{T}}$ is second countable.*

Proof. Since M is a topological manifold, it is second countable. Hence, there exists a countable basis $\mathcal{B}$ for M . Also, consider the maximal atlas $\mathcal{U} = (U_\alpha, \varphi_\alpha)_\alpha$ on M . For each coordinate open set U_α and a point $p \in U_\alpha$, there exists a basic open set $B_{p,\alpha} \in \mathcal{B}$ such that

$$p \in B_{p,\alpha} \subset U_\alpha,$$

according to the definition of bases. The collection $\{B_{p,\alpha}\}$, without duplicate elements, is a subcollection of $\mathcal{B}$, and hence, it is countable. In addition, $\{B_{p,\alpha}\}$ is also an open covering of M , that is

$$\begin{aligned} M &= \bigcup_{p \in M} \{p\} \subset \bigcup_\alpha \bigcup_{p \in U_\alpha} B_{p,\alpha} \subset \bigcup_\alpha U_\alpha = M \\ \Rightarrow \quad M &= \bigcup_\alpha \bigcup_{p \in U_\alpha} B_{p,\alpha}. \end{aligned}$$

Next, we rename all of the sets in $\{B_{p,\alpha}\}$ so that $\{B_{p,\alpha}\} = \{B_i\}_{i=1}^\infty$. For each index i , there exists a coordinate open set U_i such that $B_i \subset U_i$. The family $\{U_i\}_{i=1}^\infty$ is also an open covering of the manifold M because

$$M = \bigcup_{i=1}^\infty B_i \subset \bigcup_{i=1}^\infty U_i \subset M \quad \Rightarrow \quad M = \bigcup_{i=1}^\infty U_i.$$

Let φ_i be the local coordinate function on U_i , for any integer $i \geq 1$. Observe that $(U_i, \varphi_i)_{i=1}^\infty$ is a smooth atlas on M and a subset of the maximal atlas $\mathcal{U}$. Then, we define $\tilde{U}_i = \pi^{-1}(U_i)$ and the corresponding coordinate function $\tilde{\varphi}_i : \tilde{U}_i \rightarrow \mathbb{R}^{2n}$. Since the set

$$\tilde{\varphi}_i(\tilde{U}_i) = \varphi_i(U_i) \times \mathbb{R}^n \subset \mathbb{R}^{2n}$$

is an open subset of $\mathbb{R}^{2n}$, the topological space $\tilde{\varphi}_i(\tilde{U}_i)$ with the subspace topology is second countable (every open set of $\mathbb{R}^{2n}$ is a topological manifold). Thus, there exists a countable basis $\{D_{i,j}\}_{j=1}^\infty$ of $\tilde{\varphi}_i(\tilde{U}_i)$. Observe also that the sets $D_{i,j}$ are open in $\mathbb{R}^{2n}$ because $\tilde{\varphi}_i(\tilde{U}_i)$ is open in $\mathbb{R}^{2n}$. We claim that $\{\tilde{B}_{i,j}\}_{i,j=1}^\infty$, where $\tilde{B}_{i,j} = \tilde{\varphi}_i^{-1}(D_{i,j})$, is a countable basis of $T(M)$. Indeed, consider an arbitrary open subset A of $T(M)$. Then, we have

$$\bigcup_{i=1}^\infty A \cap \tilde{U}_i = A \cap \left(\bigcup_{i=1}^\infty \tilde{U}_i \right) = A \cap \pi^{-1} \left(\bigcup_{i=1}^\infty U_i \right) = A \cap \pi^{-1}(M) = A \cap T(M) = A.$$

In addition, $\tilde{\varphi}_i(A \cap \tilde{U}_i)$ is open in $\mathbb{R}^{2n}$ and hence, it is an open subset of $\tilde{\varphi}_i(\tilde{U}_i)$, for every integer $i \geq 1$. Therefore, fixing any point $X \in A$, there exists $\tilde{U}_i$ for some integer $i \geq 1$ such that $X \in A \cap \tilde{U}_i$. Furthermore, there exists a basic open set $D_{i,j}$ such that $\tilde{\varphi}_i(X) \in D_{i,j} \subset \tilde{\varphi}_i(A \cap \tilde{U}_i)$. As a result, we have

$$X \in \tilde{\varphi}_i^{-1}(D_{i,j}) = \tilde{B}_{i,j} \subset A \cap \tilde{U}_i \subset A.$$

Therefore, the tangent bundle $T(M)$ is second countable. $\square$

Proposition 5. *The tangent bundle $T(M)$ with the topology $\tilde{\mathcal{T}}$ is Hausdorff.*

Proof. Let X and Y be any two distinct points in $T(M)$. There are two cases that can arise. Firstly, suppose that X and Y have different projections onto M , that is, $p = \pi(X) \neq \pi(Y) = q$. Since the manifold M is Hausdorff, there exist two open sets $U \in \mathcal{N}_p$ and $V \in \mathcal{N}_q$ such that $U \cap V = \emptyset$. The corresponding sets $\tilde{U} = \pi^{-1}(U) \in \mathcal{N}_X$ and $\tilde{V} = \pi^{-1}(V) \in \mathcal{N}_Y$ are open in $T(M)$ because the set

$$\tilde{\varphi}_\alpha(\tilde{U} \cap \tilde{U}_\alpha) = \tilde{\varphi}_\alpha(\pi^{-1}(U \cap U_\alpha)) = \varphi_\alpha(U \cap U_\alpha) \times \mathbb{R}^n$$

is open in $\mathbb{R}^{2n}$ for any α , and similarly, $\tilde{\varphi}_\alpha(\tilde{V} \cap \tilde{U}_\alpha)$ is open in $\mathbb{R}^{2n}$ for any α . Furthermore, $\tilde{U}$ and $\tilde{V}$ are disjoint because

$$\tilde{U} \cap \tilde{V} = \pi^{-1}(U \cap V) = \pi^{-1}(\emptyset) = \emptyset.$$

Secondly, suppose that X and Y belong to the same tangent space $T_p(M)$, and that $p \in U_\alpha$ for some α . Since X and Y are distinct, their coordinates are different, i.e.

$$\begin{aligned} \tilde{\varphi}_\alpha(X) &= (\varphi_\alpha(p), \alpha^1, \dots, \alpha^n), \\ \tilde{\varphi}_\alpha(Y) &= (\varphi_\alpha(p), \beta^1, \dots, \beta^n), \end{aligned}$$

for two different sets of real numbers $(\alpha^1, \dots, \alpha^n)$ and $(\beta^1, \dots, \beta^n)$. Since the open subset $\tilde{\varphi}_\alpha(\tilde{U}_\alpha)$ of $\mathbb{R}^{2n}$ is Hausdorff, there exist two disjoint open subsets A and B of $\tilde{\varphi}_\alpha(\tilde{U}_\alpha)$ such that $\tilde{\varphi}_\alpha(X) \in A$ and $\tilde{\varphi}_\alpha(Y) \in B$. Clearly, the two sets $\tilde{A} = \tilde{\varphi}_\alpha^{-1}(A)$ and $\tilde{B} = \tilde{\varphi}_\alpha^{-1}(B)$ are open in $T(M)$. Also, $\tilde{A}$ and $\tilde{B}$ are disjoint because

$$\tilde{A} \cap \tilde{B} = \tilde{\varphi}_\alpha^{-1}(A \cap B) = \tilde{\varphi}_\alpha^{-1}(\emptyset) = \emptyset.$$

As a result, the tangent bundle $T(M)$ is Hausdorff. $\square$

Proposition 6. *The tangent bundle $T(M)$ of a n -dimensional smooth manifold M is a $2n$ -dimensional smooth manifold.*

Proof. From Proposition 4 and 5, the topological space $(T(M), \tilde{\mathcal{T}})$ is second countable and Hausdorff. From the construction of the topology $\tilde{\mathcal{T}}$, each map $\tilde{\varphi}_\alpha : \tilde{U}_\alpha \rightarrow \mathbb{R}^{2n}$ is bi-continuous. In addition, the collection of open sets $\{\tilde{U}_\alpha\}_\alpha$ covers the tangent bundle $T(M)$. Furthermore, any two maps $\tilde{\varphi}_\alpha$ and $\tilde{\varphi}_\beta$ are C^∞ -compatible, by Lemma 1. Therefore, $(\tilde{U}_\alpha, \tilde{\varphi}_\alpha)_\alpha$ is a smooth atlas on $T(M)$, which makes $T(M)$ a $2n$ -dimensional smooth manifold. $\square$

Remark. The object $(T(M), \pi, M)$ is an example of [vector bundles](#). In this context, M and $T(M)$ are called the *base space* and the *total space*, respectively. Also, the natural projection $\pi : T(M) \rightarrow M$ is a continuous surjective map. The fiber $\pi^{-1}(p)$ for each $p \in M$ is a vector space. Furthermore, the total space $T(M)$ locally looks like $M \times \mathbb{R}^n$. We will not go further into the details of this concept in this report.

2 Cotangent bundle

For each point $p \in M$, the tangent space $T_p(M)$ has the corresponding dual space or *cotangent space* $T_p^*(M)$, which contains all linear functionals or *covector* $\phi_p : T_p(M) \rightarrow \mathbb{R}$. In addition, the cotangent space $T_p^*(M)$ is also a vector space. Given a coordinate frame $\{E_{i,p}\}_{i=1}^n$, we have for every tangent vector $X_p \in T_p(M)$,

$$\phi_p(X_p) = \phi_p \left(\sum_{i=1}^n \alpha^i E_{i,p} \right) = \sum_{i=1}^n \alpha^i \phi_p(E_{i,p}).$$

Hence, the action of ϕ_p on each vector in the coordinate frame $\{E_{i,p}\}_{i=1}^n$ determines the action of ϕ_p on every vectors in $T(M)$. Consider a set of covector $\{\zeta_{j,p}\}_{j=1}^n$ satisfying $\zeta_{j,p}(E_{k,p}) = \delta_{jk}$. For every covector $\phi_p \in T_p^*(M)$ and tangent vector $X_p \in T_p(M)$, we have

$$\begin{aligned} \phi_p(X_p) &= \sum_{i=1}^n \alpha^i \phi_p(E_{i,p}) = \sum_{i=1}^n \sum_{j=1}^n \alpha^i \delta_{ji} \phi_p(E_{j,p}) = \sum_{i=1}^n \sum_{j=1}^n \alpha^i \zeta_{j,p}(E_{i,p}) \phi_p(E_{j,p}) \\ &= \sum_{j=1}^n \phi_p(E_{j,p}) \zeta_{j,p} \left(\sum_{i=1}^n \alpha^i E_{i,p} \right) \\ &= \sum_{j=1}^n \phi_p(E_{j,p}) \zeta_{j,p}(X_p). \end{aligned}$$

Thus, we have for every covector $\phi_p \in T_p^*(M)$,

$$\phi_p = \sum_{j=1}^n \phi_p(E_{j,p}) \zeta_{j,p}.$$

Therefore, the covectors $\zeta_{1,p}, \dots, \zeta_{n,p}$ span the cotangent space $T_p^*(M)$ and it is not difficult to show that they are also linearly independent, which makes them a dual basis. However, we want to define them in terms of local coordinate systems on M . Indeed, if the coordinate frame $\{E_{i,p}\}_{i=1}^n$ is generated by the local coordinate function $\varphi : U \rightarrow \mathbb{R}^n$ for some coordinate open set U containing p , then for $1 \leq j \leq n$, let $\zeta_{j,p} = (d\varphi^j)_p$, which is given by

$$(d\varphi^j)_p(X_p) := X_p(\varphi^j),$$

for every tangent vector $X_p \in T_p(M)$. Also, we have for $1 \leq i, j \leq n$,

$$(d\varphi^j)_p(E_{i,p}) = E_{i,p}(\varphi^j) = \left[(\varphi^{-1})_* \left(\frac{\partial}{\partial x^i} \Big|_{\varphi(p)} \right) \right] (\varphi^j) = \frac{\partial}{\partial x^i} (\varphi^j \circ \varphi^{-1}) \Big|_{\varphi(p)} = \frac{\partial x^j}{\partial x^i} \Big|_{\varphi(p)} = \delta_{ji}.$$

Since φ^j can be identified with x^j , we usually write $(d\varphi^j)_p = (dx^j)_p$. Hence, we have for every covector $\phi_p \in T_p^*(M)$,

$$\phi_p = \sum_{j=1}^n \phi_p(E_{j,p})(dx^j)_p,$$

which shows that $T_p^*(M)$ is a n -dimensional vector space. The set $\{(dx^j)_p\}_{j=1}^n$ is called the coordinate coframe.

Now, we want to gather all cotangent spaces on the manifold M into an object called the *cotangent bundle*, which is defined by

$$T^*(M) := \bigsqcup_{p \in M} T_p^*(M).$$

As with the tangent bundle, we want to construct a differential structure on $T^*(M)$. This process can be done in a similar way. First, there is a natural projection $\hat{\pi} : T^*(M) \rightarrow M$, which is given by $\hat{\pi}(\phi_p) = p$ for any covector $\phi_p \in T^*(M)$. Consider a maximal smooth atlas $\mathcal{U} = (U_\alpha, \varphi_\alpha)_\alpha$ on the manifold M . For each α , we define a subset $\hat{U}_\alpha = (\hat{\pi})^{-1}(U_\alpha)$ of the cotangent bundle $T^*(M)$ and a map $\hat{\varphi}_\alpha : \hat{U}_\alpha \rightarrow \mathbb{R}^{2n}$, which is given by

$$\hat{\varphi}_\alpha(\phi) = (\varphi_\alpha \circ \hat{\pi}(\phi), \phi(E_{1,p}), \dots, \phi(E_{n,p})) = (x^1, \dots, x^n, a_1, \dots, a_n),$$

for any covector $\phi \in \hat{U}_\alpha$ with the projection $\hat{\pi}(\phi) = p \in U_\alpha$. This map is a bijection with an inverse $\hat{\varphi}_\alpha^{-1} : \hat{\varphi}_\alpha(\hat{U}_\alpha) \rightarrow \hat{U}_\alpha$ given by

$$\hat{\varphi}_\alpha^{-1}(x^1, \dots, x^n, a_1, \dots, a_n) = \sum_{j=1}^n a_j (dx^j)_{\varphi_\alpha^{-1}(x^1, \dots, x^n)},$$

for any $(x^1, \dots, x^n, a_1, \dots, a_n) \in \hat{\varphi}_\alpha(\hat{U}_\alpha) \subset \mathbb{R}^{2n}$. The collection $(\hat{U}_\alpha, \hat{\varphi}_\alpha)_\alpha$ will later become the smooth atlas on the cotangent bundle $T^*(M)$. For now, the following lemma shows that each pair of maps $\hat{\varphi}_\alpha$ and $\hat{\varphi}_\beta$ are C^∞ -compatible with each other.

Lemma 7. *If the pair $(\hat{U}_\alpha, \hat{\varphi}_\alpha)$ and $(\hat{U}_\beta, \hat{\varphi}_\beta)$ corresponds to the two coordinate systems $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) , then the two maps*

$$\hat{\varphi}_\alpha \circ \hat{\varphi}_\beta^{-1} : \hat{\varphi}_\beta(\hat{U}_\alpha \cap \hat{U}_\beta) \rightarrow \hat{\varphi}_\alpha(\hat{U}_\alpha \cap \hat{U}_\beta),$$

and

$$\hat{\varphi}_\beta \circ \hat{\varphi}_\alpha^{-1} : \hat{\varphi}_\alpha(\hat{U}_\alpha \cap \hat{U}_\beta) \rightarrow \hat{\varphi}_\beta(\hat{U}_\alpha \cap \hat{U}_\beta)$$

are smooth maps.

Proof. First, consider any point $p \in U_\alpha \cap U_\beta$ and any tangent vector $X_p \in T_p(M)$. Then, we have for $1 \leq j \leq n$,

$$\begin{aligned} (d\varphi_\beta^j)_p(X_p) &= \sum_{k=1}^n \alpha^k (d\varphi_\beta^j)_p(E_{k,p}^\alpha) = \sum_{k=1}^n \alpha^k \frac{\partial}{\partial x^k} (\varphi_\beta \circ \varphi_\alpha^{-1})^j \Big|_{\varphi_\alpha(p)} \\ &= \sum_{k=1}^n \sum_{i=1}^n \alpha^i \frac{\partial}{\partial x^k} (\varphi_\beta \circ \varphi_\alpha^{-1})^j \Big|_{\varphi_\alpha(p)} \delta_{ki} \\ &= \sum_{k=1}^n \sum_{i=1}^n \alpha^i \frac{\partial}{\partial x^k} (\varphi_\beta \circ \varphi_\alpha^{-1})^j \Big|_{\varphi_\alpha(p)} (d\varphi_\alpha^k)_p(E_{i,p}^\alpha) \\ &= \sum_{k=1}^n \frac{\partial}{\partial x^k} (\varphi_\beta \circ \varphi_\alpha^{-1})^j \Big|_{\varphi_\alpha(p)} (d\varphi_\alpha^k)_p \left(\sum_{i=1}^n \alpha^i E_{i,p}^\alpha \right) \\ &= \sum_{k=1}^n \frac{\partial}{\partial x^k} (\varphi_\beta \circ \varphi_\alpha^{-1})^j \Big|_{\varphi_\alpha(p)} (d\varphi_\alpha^k)_p(X_p), \end{aligned}$$

where

$$\begin{aligned} (d\varphi_\beta^j)_p (E_{k,p}^\alpha) &= E_{k,p}^\alpha(\varphi_\beta^j) = \left[(\varphi_\alpha^{-1})_* \left(\frac{\partial}{\partial x^k} \Big|_{\varphi_\alpha(p)} \right) \right] (\varphi_\beta^j) = \frac{\partial}{\partial x^k} (\varphi_\beta^j \circ \varphi_\alpha^{-1}) \Big|_{\varphi_\alpha(p)} \\ &= \frac{\partial}{\partial x^k} (\varphi_\beta \circ \varphi_\alpha^{-1})^j \Big|_{\varphi_\alpha(p)}. \end{aligned}$$

Hence,

$$(d\varphi_\beta^j)_p = \sum_{k=1}^n \frac{\partial}{\partial x^k} (\varphi_\beta \circ \varphi_\alpha^{-1})^j \Big|_{\varphi_\alpha(p)} (d\varphi_\alpha^k)_p.$$

Thus, for any $(b_1, \dots, b_n) \in \mathbb{R}^n$, we have

$$\begin{aligned} \hat{\varphi}_\alpha \circ \hat{\varphi}_\beta^{-1} (\varphi_\beta(p), b^1, \dots, b^n) &= \hat{\varphi}_\alpha \left(\sum_{j=1}^n b_j (d\varphi_\beta^j)_p \right) = \hat{\varphi}_\alpha \left(\sum_{k=1}^n a_k (d\varphi_\alpha^k)_p \right) \\ &= (\varphi_\alpha \circ \varphi_\beta^{-1} (\varphi_\beta(p)), a_1, \dots, a_n), \end{aligned}$$

where

$$a_k = \sum_{j=1}^n b_j \frac{\partial}{\partial x^k} (\varphi_\beta \circ \varphi_\alpha^{-1})^j \Big|_{\varphi_\alpha(p)}.$$

Therefore, since $\varphi_\alpha \circ \varphi_\beta^{-1}$ and its inverse $\varphi_\beta \circ \varphi_\alpha^{-1}$ are smooth maps, the map $\hat{\varphi}_\alpha \circ \hat{\varphi}_\beta^{-1}$ is also a smooth map. A similar calculation can also be done for the map $\hat{\varphi}_\beta \circ \hat{\varphi}_\alpha^{-1}$. $\square$

The topology on the cotangent bundle $T^*(M)$ can be constructed in the same way as the tangent bundle. A basis for a topology on $T^*(M)$ consists of all subsets A such that $A \subset \hat{U}_\alpha$ and $\hat{\varphi}_\alpha(A)$ is an open subset of $\mathbb{R}^{2n}$. This collection of subsets of $T^*(M)$ does indeed satisfy the two conditions in Proposition A3, which can be proved by employing similar set-theoretic arguments used in the proof of Lemma 2. Then, the cotangent bundle $T^*(M)$ with the topology generated by this basis is a second countable, Hausdorff topological space, which can also be proved by using similar arguments in the proofs of Proposition 4 and 5. In addition, each map $\hat{\varphi}_\alpha : \hat{U}_\alpha \rightarrow \mathbb{R}^{2n}$ is a homeomorphism from $\hat{U}_\alpha$ to $\hat{\varphi}_\alpha(\hat{U}_\alpha)$ due to the construction of the topology on $T^*(M)$. Also, $\{\hat{U}_\alpha\}$ is an open covering of $T^*(M)$ because $\hat{\varphi}_\alpha(\hat{U}_\alpha) = \varphi_\alpha(U_\alpha) \times \mathbb{R}^n$ is an open subset of $\mathbb{R}^{2n}$ ($\implies \hat{U}_\alpha$ is open in $T^*(M)$), and

$$T^*(M) = \hat{\pi}^{-1}(M) = \hat{\pi}^{-1} \left(\bigcup_\alpha U_\alpha \right) = \bigcup_\alpha \hat{\pi}^{-1}(U_\alpha).$$

Along with the C^∞ compatibility between any two maps $\hat{\varphi}_\alpha$ and $\hat{\varphi}_\beta$ (Lemma 7), this shows that $\hat{\mathcal{U}} = (\hat{U}_\alpha, \hat{\varphi}_\alpha)$ is a smooth atlas on $T^*(M)$, which implies that the manifold $T^*(M)$ with the atlas $\hat{\mathcal{U}}$ is a $2n$ -dimensional smooth manifold.

Remark. The object $(T^*(M), \hat{\pi}, M)$ is also an example of vector bundles.

Appendices

A Bases of a topology

Definition A1 (Basis). A subset $\mathcal{B} := \{V_\alpha\} \subset \mathcal{T}$ is a basis for the topological space $(M, \mathcal{T})$ if for any $p \in M$ and any $U \in \mathcal{N}_p$, there exists $V \in \mathcal{B}$ such that $p \in V \subset U$.

Lemma A2. *In a topological space $(M, \mathcal{T})$, a collection $\mathcal{B}$ of open sets of M is a basis if and only if every open set in M is a union of sets in $\mathcal{B}$.*

Proof. First, suppose that $\mathcal{B}$ is a basis and U is an open set in M . For any point $p \in U$, there exists a basic open set $B_p \in \mathcal{B}$ such that $p \in B_p \subset U$. Then,

$$U = \bigcup_{p \in U} \{p\} \subset \bigcup_{p \in U} B_p.$$

On the other hand,

$$B_p \subset U, \forall p \in U \Rightarrow \bigcup_{p \in U} B_p \subset U.$$

Therefore, $U = \bigcup_{p \in U} B_p$.

Conversely, suppose that every open set in M is a union of sets in $\mathcal{B}$. Hence, for any open set U , it can be written in the form $U = \bigcup_\alpha B_\alpha$ for some family $\{B_\alpha\}_\alpha \subset \mathcal{B}$. Given any point $p \in U$, there exists a set $B_\beta \in \{B_\alpha\}_\alpha \subset \mathcal{B}$ such that $p \in B_\beta \subset U$. As a result, $\mathcal{B}$ is a basis. $\square$

The following proposition provides conditions for a collection of subsets to be a basis for some topology and how to construct a topology from this basis.

Proposition A3. *A collection $\mathcal{B}$ of subsets of a set M is a basis for some topology $\mathcal{T}$ on M if and only if*

- i) M is the union of all sets in $\mathcal{B}$,
- ii) given any two sets B_1 and B_2 in $\mathcal{B}$ and $p \in B_1 \cap B_2$, there is a set $B \in \mathcal{B}$ such that $p \in B \subset B_1 \cap B_2$.

Proof. Suppose that a collection $\mathcal{B}$ of subsets is a basis for some topology $\mathcal{T}$ of M . Since $M \in \mathcal{T}$, by Lemma A2, the set M is a union of sets in $\mathcal{B}$, i.e. $M = \bigcup_\alpha B_\alpha$ for some family $\{B_\alpha\}_\alpha \subset \mathcal{B}$. Furthermore, since

$$M = \bigcup_\alpha B_\alpha \subset \bigcup_{B \in \mathcal{B}} B,$$

and

$$B \subset M, \forall B \in \mathcal{B} \Rightarrow \bigcup_{B \in \mathcal{B}} B \subset M,$$

the set M is the union of all the sets in $\mathcal{B}$, i.e. $M = \bigcup_{B \in \mathcal{B}} B$ ($\Rightarrow$ condition (i)).

Since $\mathcal{B}$ is a subset of the topology $\mathcal{T}$, any two sets $B_1, B_2 \in \mathcal{B}$ are also open sets, and so is the intersection $B_1 \cap B_2$. By the definition A1 of bases, given any point $p \in B_1 \cap B_2$, there exists a set $B \in \mathcal{B}$ such that $p \in B \subset B_1 \cap B_2$ ($\Rightarrow$ condition (ii)).

Conversely, suppose that a collection $\mathcal{B}$ of subsets of a set M satisfies the conditions (i) and (ii). Define $\mathcal{T}$ to be the collection of all sets that are unions of sets in $\mathcal{B}$. Then, $M \in \mathcal{T}$ due to the condition (i). Also, $\emptyset \in \mathcal{T}$ because $\emptyset = \bigcup_{\alpha \in \emptyset} B_\alpha$ for $B_\alpha \in \mathcal{B}$. By the construction of $\mathcal{B}$,

arbitrary unions of sets in $\mathcal{B}$ are also in $\mathcal{B}$. For showing that every finite intersection of sets in $\mathcal{B}$ still belongs to $\mathcal{B}$, take any two sets $U = \bigcup_{\alpha} B_{\alpha}$ and $V = \bigcup_{\beta} B_{\beta}$ in $\mathcal{T}$. Then,

$$\begin{aligned} U \cap V &= \left(\bigcup_{\alpha} B_{\alpha} \right) \cap \left(\bigcup_{\beta} B_{\beta} \right) \\ &= \bigcup_{\alpha, \beta} B_{\alpha} \cap B_{\beta}. \end{aligned}$$

This means that any point $p \in U \cap V$ also belongs to $B_{\alpha} \cap B_{\beta}$ for some α and β . By the condition (ii), there exists a set $B_p \in \mathcal{B}$ such that $p \in B_p \subset B_{\alpha} \cap B_{\beta}$. Since

$$U \cap V = \bigcup_{p \in U \cap V} \{p\} \subset \bigcup_{p \in U \cap V} B_p,$$

and

$$B_p \subset B_{\alpha} \cap B_{\beta} \subset U \cap V, \forall p \in U \cap V \Rightarrow \bigcup_{p \in U \cap V} B_p \subset U \cap V,$$

the intersection $U \cap V$ is also an element of $\mathcal{T}$, i.e. $U \cap V = \bigcup_{p \in U \cap V} B_p \in \mathcal{T}$. By induction, any finite intersection of sets in the collection $\mathcal{T}$ are also sets in the collection $\mathcal{T}$. Therefore, $\mathcal{T}$ is a topology of M .

□

References

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